GOPIKRISHNA NALLAGORLA

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EXPERIENCE

Elai AgriTech Pvt Ltd — *Pune, Maharashtra*

Data Scientist | Core Founding Team Member **Dec 2023 – Present**

- Joined as a **core founding team member** and built Elai AgriTech's entire remote sensing and machine learning function from the ground up — from infrastructure and pipelines to production-grade AI systems powering the company's core agricultural intelligence capabilities.
- Engineered ensemble ML (**Random Forest, XGBoost**) and deep learning (**LSTM, CNN**) models on multi-temporal satellite and weather data, achieving **90% crop classification accuracy** across 15+ crop types and **87% yield prediction accuracy** 3–4 months ahead of harvest.
- Designed and deployed a **CNN-based image segmentation** system for automated farm boundary delineation — handling geometric, radiometric, and atmospheric correction plus cloud masking — achieving **80% boundary precision** across diverse agricultural landscapes.
- Led the end-to-end design and deployment of an AI-powered **pest and disease identification system** covering **300+ classes** (MobileNet, ResNet; >85% accuracy), with ONNX export, **Grad-CAM** explainability, and a **Llama-3** conversational diagnostic agent — directly securing **internal R&D funding**.
- Built automated satellite data pipelines integrating **Google Earth Engine** and Python automation, reducing manual analysis effort by **70%** and enabling the team to operate at significantly greater scale.
- Developed **real-time crop health monitoring** systems and interactive dashboards, translating complex geospatial outputs into actionable, stakeholder-ready insights.
- Architected the **Elai Agri Chatbot**, a conversational AI system translating NDVI scores, crop health reports, and satellite assessments into plain-language guidance for farmers, agronomists, and loan officers.
- Pioneered a **semi-automated AI data collection agent** for building large-scale, balanced pest/disease image datasets via multi-source web scraping, reducing dataset preparation time by **70%**.
- Deployed automated **anomaly detection** systems for early identification of crop stress, disease outbreaks, and irrigation failures, contributing to an estimated **25% reduction in crop loss risk**.
- Established the company's technical SOPs for geospatial data processing and quality control; led and mentored multi-disciplinary teams across field operations, data processing, and analytics.

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, JavaScript, OOP, Functional Programming
- **Geospatial & Remote Sensing:** Google Earth Engine (GEE), QGIS, ArcGIS, GDAL, Rasterio, GeoPandas, Shapely, PostGIS
- **Satellite Data & Vegetation Indices:** Sentinel-1/2, Landsat 8/9, MODIS, NASA POWER; NDVI, EVI, SAVI, NDRE, NDMI, NBR, PSRI, MSAVI, SIPI, GCI
- **Machine Learning & AI:** Scikit-learn, XGBoost, LightGBM, Random Forest, Isolation Forest, SVM, KNN, TensorFlow, Keras, PyTorch
- **Deep Learning:** CNN, RNN, LSTM, Autoencoders, Transfer Learning, MobileNet/ResNet
- **Explainable AI:** SHAP, Grad-CAM
- **LLMs & AI Agents:** Groq SDK, Llama-3, Ollama, Prompt Engineering, Sarvam API (multilingual), AI Data Collection Agents
- **Deployment & MLOps:** FastAPI, Flask, Docker/Docker Compose, Celery, ONNX Runtime, GitHub Actions CI/CD, Render/Netlify
- **Cloud & Databases:** AWS (S3, EC2, SageMaker, Lambda), GCP, MongoDB, Redis, PostgreSQL, Supabase, MinIO/Cloudflare R2

PROJECTS

Enhanced Satellite Crop Monitoring & Decision-Support Pipeline — *Production, Elai AgriTech*

- Architected and shipped a production **11-stage automated crop monitoring pipeline** (FastAPI + CLI) fusing Sentinel-2 optical, Sentinel-1 SAR, and Open-Meteo weather data across **16 crops** on a ~6-day cadence.
- Built multi-sensor satellite ingestion computing **20 vegetation indices** server-side plus SAR-based soil-moisture inversion (Water Cloud Model) for all-weather monitoring through India's cloud-heavy kharif season.
- Developed cloud-gap imputation, a biomass accumulation model (RUE/fAPAR/LAI), multi-type stress classification, a pest/disease risk engine (~152 agent rules), yield projection, and soil nutrient proxy models.
- Deployed as a containerized **FastAPI** service (**Docker**, ECS Fargate) with an async job queue, Redis-backed store, and MongoDB-integrated GeoTIFF dashboards for real-time, farm-level decision support.

Elai Agri Chatbot — *Internal Deployment, Elai AgriTech*

- Architected a complete conversational AI system bridging complex remote sensing outputs and non-technical stakeholders — farmers, agronomists, and loan officers.
- Translated NDVI scores, crop health reports, and satellite assessments into plain-language, actionable guidance, improving stakeholder comprehension and decision-making speed.

Multi-Crop Classification & Yield Prediction System — *Production, Elai AgriTech*

- Engineered ensemble ML (**Random Forest, XGBoost**) and deep learning (**LSTM**) models on multi-temporal NDVI, weather, and soil data for crop classification and yield forecasting.
- Achieved **90% crop classification accuracy** across 15+ crop types and **87% yield prediction accuracy** delivered 3–4 months ahead of harvest.

Pest & Disease Identification System — *Production, Elai AgriTech*

- Built a deep learning pipeline (**MobileNet, ResNet**) covering **300+ pest and disease classes** with ONNX export for lightweight production deployment, achieving **>85% accuracy**.
- Integrated **Grad-CAM** explainable AI heatmaps and a **Llama-3** conversational diagnostic agent for natural-language recommendations; deployed via FastAPI, Supabase, PostgreSQL, Docker, and GitHub Actions CI/CD on Render.
- Directly secured **internal R&D funding**, validating the platform's AI capability to stakeholders.

Satellite-Based Carbon MRV Platform — *Independent Project*

- Designed and built a full-stack, **IPCC/Verra (VM0042/VM0047)-compliant** carbon MRV platform from scratch — satellite ingestion (GEE), multi-pool carbon calculation (AGB, BGB, SOC, Deadwood, Litter), and an uncertainty engine with conservative deduction buffers.
- Implemented **LightGBM + SHAP** biomass/SOC estimation, automated audit-ready PDF reporting, and Groq-powered (Llama 3.3 70B) LLM insights, with full async processing via Celery/Redis and Docker Compose deployment (FastAPI, MongoDB, Next.js).

Satellite-Based Farm Credit Scoring System — *Independent Project*

- Built a **12-stage fully automated pipeline** scoring smallholder farmer creditworthiness from satellite data alone — continuous Sentinel-2 monitoring, crop cycle detection, and hybrid credit scoring (rule-based + Isolation Forest + XGBoost/RF).
- Delivered AI-generated, multilingual (Sarvam API) farmer credit reports with a 0–1000 risk score and credit-limit recommendations, enabling bank-grade lending decisions with **zero field visits**.

Regional Agricultural Intelligence Platform — *Independent Project*

- Architected a real-time satellite monitoring platform for government and enterprise use, delivering crop health, drought, and land-use intelligence at regional to national scale.
- Built async GEE pipelines (Celery), multi-temporal Sentinel/Landsat analysis, and interactive decision-maker dashboards (FastAPI, MongoDB, Next.js, Docker) with isolated multi-product access control.

POSITIONS OF RESPONSIBILITY

Core Team Member & Project Lead — Elai AgriTech Pvt Ltd

- Key founding team member instrumental in building the company's geospatial and remote sensing capabilities, establishing technical frameworks and operational workflows.

- Led and organized multi-disciplinary teams for complex remote sensing projects, coordinating between field operations, data processing, and analytical teams.
- Managed end-to-end project execution for major agricultural mapping initiatives, ensuring quality deliverables and adherence to project timelines.
- Established standard operating procedures for geospatial data processing, quality control protocols, and technical documentation standards.
- Mentored team members on advanced remote sensing techniques and GIS methodologies, building internal technical capacity.

EDUCATION

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT)
M.Sc. Data Analytics (Agriculture Analytics) **2022 – 2024**

- CGPA: **7.1**

Dr. Panjabrao Deshmukh Krishi Vidyapeeth (Agriculture University)
B.Sc. (Hons.) Agriculture **2018 – 2022**

- CGPA: **7.67**

PROFESSIONAL SKILLS

Spatial Problem Solving • Technical Communication • Project Management • Field Survey Coordination • Cross-Functional Collaboration

INTERESTS

Geospatial Technology Innovation • Agricultural Sustainability • Outdoor Activities and Sports • Technical Blogging on Remote Sensing Applications

LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Telugu (Native)